PART 1 - GENERAL

23 81 23 Part 1.1.1.A This section specifies the requirements for in-row cooling / crah for the Meridian Data Centre Campus - Phase 1.

1.1 Summary

23 81 23 Part 1.1.1.B The scope of supply includes all equipment, accessories, and installation materials required for a complete and operational system.

23 81 23 Part 1.2.1.A The following standards shall apply: ASHRAE TC 9.9, AHRI 1360 and all applicable local codes.

1.2 References

23 81 23 Part 1.3.1 The Contractor shall submit the equipment submittal package referencing the current active project specification revision Rev 2024. Submittals referencing an incorrect or outdated specification revision shall be returned without review.

1.3 Submittal Requirements

23 81 23 Part 1.3.2.A Submittals shall include certified manufacturer product data sheets, performance test reports, dimensional drawings, and a clause-by-clause compliance matrix.

23 81 23 Part 1.4.1.A Manufacturer shall have a minimum of 10 years experience in the design and manufacture of in-row cooling / crah, with at least 5 reference installations in Tier III or above data centres.

1.4 Quality Assurance

23 81 23 Part 1.4.1.B Manufacturing facility shall hold current ISO 9001 and ISO 14001 certifications.

23 81 23 Part 1.5.1.A The manufacturer shall provide a minimum of 24 months on-site warranty from the date of commissioning.

1.5 Warranty and Spares

23 81 23 Part 1.5.1.B A recommended spare parts list with pricing shall be submitted for Consultant review.

PART 2 - PRODUCTS

2.1 Acceptable Manufacturers and System Configuration

23 81 23 Part 2.1.1.A Acceptable manufacturers: CryoCore or approved equivalent meeting all performance requirements of this section.

23 81 23 Part 2.1.2.A The system shall be configured in a fully redundant N+1 configuration.

23 81 23 Part 2.2.1.A Performance shall meet the requirements of the applicable standards cited in Part 1.2.1.A.

2.2 Performance and Design Criteria

23 81 23 Part 2.2.2.A All equipment shall be rated for continuous operation at 40°C ambient temperature.

23 81 23 Part 2.2.3.A System efficiency at full rated load shall be a minimum of 95.0% in normal operating mode.

23 81 23 Part 2.3.1.A Equipment enclosure shall be constructed of minimum 1.6 mm steel sheet with IP20 minimum ingress protection per IEC 60529.

2.3 Physical Construction

23 81 23 Part 2.3.2.A Enclosure paint finish shall be RAL 7035 (Light Grey) polyester powder coating, minimum 60 micron thickness.

23 81 23 Part 2.3.3.A Equipment shall be suitable for installation on raised access floor with standard pedestal height of 600 mm.

23 81 23 Part 2.4.1.A All internal power connections shall utilise bolted joints with calibrated torque values marked on the assembly.

PART 3 - EXECUTION

3.1 Examination and Preparation

23 81 23 Part 3.1.1.A Prior to delivery, the Contractor shall verify that the installation area is complete and ready to receive equipment.

3.2 Installation

23 81 23 Part 3.2.1.A Installation shall be per manufacturer's written instructions and approved shop drawings.

3.3 Field Quality Control

23 81 23 Part 3.3.1.A Perform insulation resistance testing on all power cables prior to energisation. Minimum acceptable insulation resistance is 100 MΩ at 1000 VDC.

3.4 Commissioning and Acceptance Testing

23 81 23 Part 3.4.1.A The commissioning agent shall perform a complete functional test of each system.

23 81 23 Part 3.4.2.A Verify performance meets or exceeds the specified values in Part 2 of this section.

23 81 23 Part 3.4.2.B Simulate system failure and verify redundant takeover. Remaining equipment shall assume full load without interruption.

23 81 23 Part 3.4.2.C Perform BMS communication integration test. Verify all alarm and status signals are correctly displayed on the central BMS operator console.

23 81 23 Part 3.4.2.D Execute a coordinated full-load transfer test from utility to standby and back. Verify uninterrupted operation throughout the sequence.